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Air Flow Measurements in the Bag

Many people in the building performance field know about a certain, easy-to-perform air flow test that involves common household items. If you are not among them, read on. If you already know about the garbage bag air flow test read on anyway—you might learn something new about it.

To perform the test, take a garbage bag, a wire coat hanger that you don't hang clothes on, some adhesive tape, and a watch. Bend the coat hanger into a circle, square, or rectangle big enough to fit completely around the register where you want to measure air flow. Fit the opening of the garbage bag around the coat hanger and tape it in place (you don't have to make it tight, just so it stays on the hanger). Now you have the necessary technology. The biggest challenge for those who are technologically inept is how to avoid poking a hole in the garbage bag with the coat hanger. The authors welcome any tips regarding this point.

Put the completely deflated garbage bag rig that you've created over a supply

register. Time how long it takes for the bag to fully inflate. Or put the fully inflated garbage bag rig over a return register and time how long it takes for the bag to deflate. While in theory you could simply divide the volume of the bag by the time it takes to inflate or deflate, most practitioners choose to use a calibration table to determine air flow (see Table 1).

The Canada Mortgage and Housing Corporation (CMHC) began publicizing the garbage bag air flow test in the early '90s and has used it in several field tests (usually when contractors objected to using CMHC's cumbersome duct test rig). In its literature, CMHC specifies using a Glad 66 cm x 91 cm garbage bag. When your bag is laid out flat, it should approximate that bag's dimensions: 26 inches by 36 inches (about 2 ft x 3 ft). Easy enough to measure, or even approximate.

The garbage bag air flow test is very appropriate for estimating forced-air furnace flows and for help in balancing HVAC systems (see "Garbage Bag Educator"). It can also be useful when installing or checking flows from

exhaust devices such as bathroom or kitchen fans. Sometimes you can raise a full bag to the exhaust grille and time its deflation. Often, it is easier to go to the outlet hood and time the bag's inflation. We recommended this method to homeowners who want to do a rough check on heat recovery ventilator (HRV) balance.

One of the first uses of the garbage bag test was for someone in Canada who was getting no heat on the second floor in his relatively new suburban house. Several tradespeople had tried to solve the problem with no success. The man's furnace was oversized, as is true in most new housing, so there should have been no shortage of heat. The garbage bag test showed that only 2 of his 18 supply ducts had flows greater than 10 CFM. His duct system was essentially blocked, probably by the air conditioning coil, since the only two registers with decent flow took off before the AC coil.

The beauty of the garbage bag test is its availability. Even if you forget to bring any equipment or the calibration

Garbage Bag Educator

I have been teaching the garbage bag air flow test to Canadian contractors, builders, and building officials in training courses, on-site, and over the phone for six or seven years, with positive results. I always carry my garbage bag kit in the truck.

Recently a production builder client who builds more than 500 homes per year was having difficulty with ventilation system inspections. According to the local code, if a home has more than four bedrooms (as is usually the case when the home has a basement apartment), the building official may

ask for a performance review of the ventilation system by a qualified person (such as myself). The performance review is done according to the Canadian Standards Association (CSA) F326 Ventilation standard, which requires physical measurement of ventilation air flow. Most of the time this comes up a day or so before the house is to be delivered. If the ventilation system doesn't pass, it's a big deal. Most of the houses were not passing—but for stupid reasons such as backdraft dampers stuck closed, a duct packed with insulation (someone forgot to remove it), and so on.

I taught the site super the garbage bag method, and this problem has been eliminated. Performance reviews now go smoothly, since any problem with the ventilation system has been eliminated before I arrive.

—Dara Bowser

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Table 1. Garbage Bag Test Calibration Tables

Small Green Garbage Bag (Glad 66 cm x 91 cm)

Time to Inflate	Air Flow
2 seconds	75 CFM (35 L/s)
4 seconds	40 CFM (20 L/s)
10 seconds	20 CFM (10 L/s)

For deflation, add 1 second.

Big Orange Garbage Bag (Glad 79 cm x 119 cm)

Time to Inflate	Air Flow
2 seconds	210 CFM (100 L/s)
4 seconds	105 CFM (50 L/s)
6 seconds	75 CFM (35 L/s)
10 seconds	40 CFM (20 L/s)

Source: Canada Mortgage and Housing Association

Note: The calibration air flows are measured values and are not derived from bag volume.

nature of the test is easy for homeowners to interpret. A 2-second inflation (75 CFM) is a healthy, robust flow. An inflation of less than 10 CFM (roughly 15 seconds) is almost comically slow. The relative performance of several supply registers is obvious even without a watch.

We would like to see the bag test frequently used by anyone who installs air-moving devices. It's not designed to replace standardized HRV balancing procedures or house testing with devices such as the Duct Blaster. However, it provides a quick and effective check after the installation of forced-air ducting. You can get a good idea of the performance of all of the ducts in a house, usually within ten minutes. It is also a quick way to do a commissioning when exhaust fans are installed in houses. It is convenient, convincing, and costs next to nothing. And it's supported by both the garbage bag and the wire hanger industries.



—Jim Gunshinan and Don Fugler

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table, you can still build a test rig and use it to measure air flow in almost any house within ten minutes. "Excuse me, have you got a garbage bag and a wire coat hanger and some tape?" The visual

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For more on the CMHC garbage bag air flow test, go to CMHC's About Your House Web site: www.cmhc-schl.gc.ca/en/burema/gesein/abhose/index.cfm. Once there, scroll down to Testing Airflow for a description of the garbage bag test. For information about performing an HRV balance test, scroll down to Maintaining Your Heat Recovery Ventilator.